



Problem Solving With C - UE25CS151B

Arrays, Initialization and Traversal

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PROBLEM SOLVING WITH C

Arrays, Initialization and Traversal



1. What is an Array?
2. Properties of Arrays.
3. Classification of Arrays
4. Declaration and Initialization
5. Representation of the Array
6. Array Traversal

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What is an Array?

- A linear data structure, which is a Finite collection of similar data items stored in successive or consecutive memory locations
- **Homogenous types of data** allowed in any array. May contain all integer or all character elements, but not both together.
- `char c_array [10]; short s_array [20];`

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Properties of Arrays

- Non-primary data type or secondary data type
- Memory allocation is contiguous in nature
- Elements need not be unique.
- Demands same /homogenous types of elements
- Random access of elements in array is possible
- Elements are accessed using index/subscript
- Index or subscript always start from 0
- Memory is allocated at compile time.
- Size of the array is fixed at compile time. Returns the number of bytes occupied by the array.
- Cannot change the size at runtime.
- Arrays are assignment incompatible.
- Accessing elements of the array outside the bound can have undefined behavior at runtime

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Classification of Arrays

Category 1:

- Fixed Length Array
 - Size of the array is fixed at compile time
- Variable Length Array – Not discussed here

Category 2:

- One Dimensional (1-D) Array
 - Stores the data elements in a single row or column.
- Multi Dimensional Array

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Declaration and Initialization

Declaration

Syntax:

Data_type Array_name[Size];

Data_type: Specifies the type of the element that will be contained in the array

Array_name: Identifier to identify a variable

Size: Indicates the max no. of elements that can be stored inside the array

Example: `double x[15];` // Can contain 15 elements of type double, 0 to 14 are valid array indices or subscript

- Subscripts in array can be integer constant or integer variable or expression that yields integer
- C performs no bound checking. Care should be taken to ensure that the array indices are within the declared limits

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Declaration and Initialization

Initialization

- After an array is declared, it must be initialized.
- An array can be initialized at either **compile time** or at **runtime**.

Compile time Initialization

- `data-type array-name[size] = { list of values };`
- `float area[5]={ 23.4, 6.8, 5.5 }; // Partial initialization`
- `int a[15] = {[2] = 29, [9] = 7, [14] = 48}; //C99's designated initializers`
- `int a[15] = {0,0,29,0,0,0,0,0,0,7,0,0,0,0,48}; //C99's designated initializers`
- `int arr[] = {2, 3, 4}; // sizeof arr is decided`
- `int marks[4]={ 67, 87, 56, 77, 59 }; // undefined behavior`
- Coding Examples

Runtime Initialization

- Using a loop and input function in C
- Coding examples

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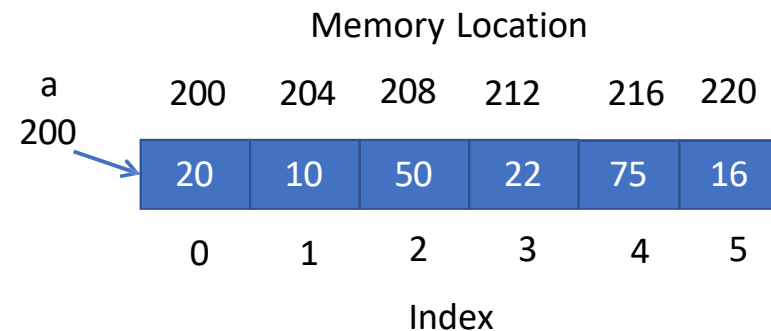
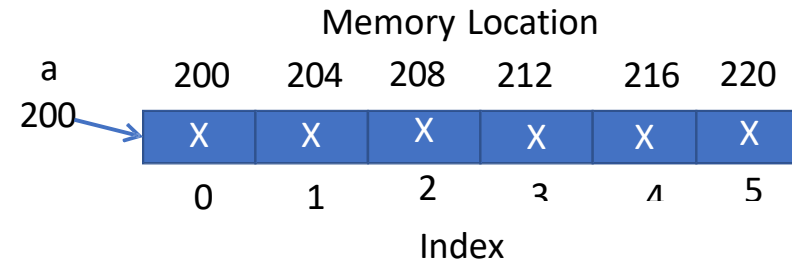
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Representation of the Array

- `int a[6];`
- `int a[6] = {20, 10, 50, 22, 75, 16};`
- Address of the first element is called the Base address of the array.
- Address of *i*th element of the array can be found using formula:

Address of *i*th element = Base address + (size of each element * *i*)



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Traversal

- Accessing each element of the array
- `int a[10];`
 - How do you access the 5th element of the array? // `a[4]`
 - How do you display each element of the array? // Using Loop
 - How much memory allocated for this?
Number of bytes allocated = size specified * size of integer
- Anytime accessing elements outside the array bound is an undefined behavior
- Coding examples



THANK YOU

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